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The Folic Acid and Creatine Trial: Treatment Effects of Supplementation on Arsenic Methylation Indices and Metabolite Concentrations in Blood in a Bangladeshi Population

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Background: Chronic arsenic (As) exposure is a global environmental health issue. Inorganic As (InAs) undergoes methylation to monomethyl (MMAs) and dimethyl-arsenical species (DMAs); full methylation to DMAs facilitates urinary excretion and is associated with reduced risk for As-related health outcomes. Nutritional factors, including folate and creatine, influence one-carbon metabolism, the biochemical pathway that provides methyl groups for As methylation.

Objective: Our aim was to investigate the effects of supplementation with folic acid (FA), creatine, or the two combined on the concentrations of As metabolites and the primary methylation index (PMI: MMAs/InAs) and secondary methylation index (SMI: DMAs/MMAs) in blood in Bangladeshi adults having a wide range of folate status.

Methods: In a randomized, double-blinded, placebo (PBO)-controlled trial, 622 participants were recruited independent of folate status and assigned to one of five treatment arms: *a*) PBO ($n = 102$), *b*) 400 μg FA/d (400FA; $n = 153$), *c*) 800 μg FA/d (800FA; $n = 151$), *d*) 3 g creatine/d (creatine; $n = 101$), or *e*) 3 g creatine + 400 μg of FA/d (creatine + 400FA; $n = 103$) for 12 wk. For the following 12 wk, half of the FA participants were randomly switched to the PBO while the other half continued FA supplementation. All participants received As-removal water filters at baseline. Blood As (bAs) metabolites were measured at weeks 0, 1, 12, and 24.

Results: At baseline, 80.3% ($n = 489$) of participants were folate sufficient (≥9 nmol/L in plasma). In all groups, bAs metabolite concentrations decreased, likely due to filter use; for example, in the PBO group, blood concentrations of MMAs (bMMAs) (geometric mean ± geometric standard deviation) decreased from 3.55 ± 1.89 μg/L at baseline to 2.73 ± 1.74 at week 1. After 1 wk, the mean within-person increase in SMI for the creatine + 400FA group was greater than that of the PBO group ($p = 0.05$). The mean percentage decrease in bMMAs between baseline and week 12 was greater for all treatment groups compared with the PBO group [400FA: −10.4 (95% CI: −11.9, −8.75), 800FA: −9.54 (95% CI: −11.1, −7.97), creatine: −5.85 (95% CI: −8.59, −3.03), creatine + 400FA: −8.44 (95% CI: −9.95, −6.90), PBO: −2.02 (95% CI: −4.03, 0.04)], and the percentage increase in blood DMAs (bDMAs) concentrations for the FA-treated groups significantly exceeded that of PBO [400FA: 12.8 (95% CI: 10.5, 15.2), 800FA: 11.3 (95% CI: 8.95, 13.8), creatine + 400FA: 7.45 (95% CI: 5.23, 9.71), PBO: −0.15 (95% CI: −2.85, 2.63)]. The mean decrease in PMI and increase in SMI in all FA groups significantly exceeded PBO ($p < 0.05$). Data from week 24 showed evidence of a reversal of treatment effects on As metabolites from week 12 in those who switched from 800FA to PBO, with significant decreases in SMI [−9.0% (95% CI: −3.5, −14.8)] and bDMAs [−5.9% (95% CI: −1.8, −10.2)], whereas PMI and bMMAs concentrations continued to decline [−7.16% (95% CI: −0.48, −14.3) and −3.1% (95% CI: −0.1, −6.2), respectively] for those who remained on 800FA supplementation.

Conclusions: FA supplementation lowered bMMAs and increased bDMAs in a sample of primarily folate-replete adults, whereas creatine supplementation lowered bMMAs. Evidence of the reversal of treatment effects on As metabolites following FA cessation suggests short-term benefits of supplementation and underscores the importance of long-term interventions, such as FA fortification. https://doi.org/10.1289/EHP11270

Introduction

Arsenic (As) contamination of groundwater is common in many regions of the world. At least 94 million people worldwide are exposed to concentrations exceeding the World Health Organization (WHO) guideline of 10 μg/L in groundwater.[1–3] In Bangladesh, where well water is used as an alternative to microbially contaminated surface water, ∼30 million people remain exposed to drinking water with high As concentrations (>10 μg/L).[4]

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Arsenic exposure affects all organ systems, and chronic As exposure is associated with numerous adverse health effects, including cardiovascular disease, diabetes and diabetes-related outcomes, skin lesions (i.e., melanosis, leukomelanosis, and keratosis), impaired intellectual function, and cancers (e.g., bladder, lung, skin cancers).[5–7] Arsenic metabolism may modify disease risk. Ingested inorganic As (InAs) undergoes a series of reduction and oxidative methylation reactions. According to the Challenger pathway, arsenite ($InAs^{III}$) is methylated to monomethylarsonic acid ($MMAs^{V}$) by arsenic-3-methyltransferase (AS3MT) using *S*-adenosylmethionine (SAM) as the methyl donor. $MMAs^{V}$ is then reduced to $MMAs^{III}$ and methylated to form dimethylarsinic acid ($DMAs^{V}$) (Figure 1).[9] The complete methylation of InAs to DMAs is important because toxicological studies in human lung and bladder cells and hepatocytes have shown that $MMAs^{III}$ is the most cytotoxic and genotoxic As species.[10,11] It is difficult to distinguish between $MMAs^{V}$ and $MMAs^{III}$ in epidemiological studies owing to oxidation in biological samples; therefore, total MMAs ($MMAs^{(III+V)}$) is reported. Among As-exposed populations, a higher proportion of $MMAs^{(III+V)}$ (and a lower proportion of DMAs) in urine has been associated with an increased risk for cancers (e.g., bladder, lung, skin cancers), skin lesions, peripheral vascular disease, and atherosclerosis.[12,13]

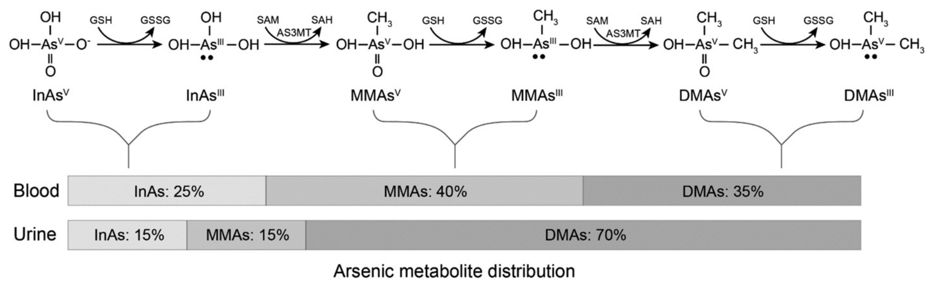

Figure 1. Relative distribution of arsenic metabolites in blood and urine. Inorganic arsenic (InAs) in the form of arsenite ($InAs^{III}$) is methylated to form monomethylarsonic acid ($MMAs^{V}$) in a reaction that is catalyzed by arsenic (+3 oxidation state) methyltransferase (AS3MT) using the methyl donor *S*-adenosylmethionine (SAM). $MMAs^{V}$ is then reduced to form methylarsonous acid ($MMAs^{III}$), and a subsequent methylation yields dimethylarsinic acid ($DMAs^{V}$) and *S*-adenosylhomocysteine (SAH). Total blood arsenic is composed of $\sim 25\%$ InAs, 40% MMAs, and 35% DMAs, whereas total urine arsenic is composed of $\sim 15\%$ InAs, 15% MMAs, and 70% DMAs.[8]

One-carbon metabolism (OCM), the metabolic pathway that generates SAM, is influenced by nutrients, including folate and creatine. Folate, in the form of 5-methyltetrahydrofolate (5-mTHF), donates a methyl group to homocysteine to form methionine, which is activated by methionine adenosyltransferase to form SAM. An estimated 40% of SAM is consumed by the methylation of guanidinoacetate (GAA) to form creatine, an amino acid derivative that is found in tissues with high energy requirements, such as muscle cells.[14] Creatine synthesis is down-regulated by dietary intake of creatine, predominantly through the consumption of meat and by creatine supplementation.[15] Increased creatine consumption may increase SAM available for the methylation of As.

In a previous randomized controlled trial (RCT), our group investigated the effect of folic acid (FA) supplementation on As methylation capacity among folate-deficient Bangladeshi adults (plasma folate concentrations $<9\,\mathrm{nmol/L}$) exposed to As through drinking water. Following 12 wk of treatment, the proportion of DMAs (%DMAs) in urine increased while %InAs and %MMAs in urine, as well as total blood As (bAs) and blood MMAs (bMMAs) concentrations decreased among participants receiving 400 μg FA/d compared with the placebo (PBO) group.[8,16]

More recently, in the Folic Acid and Creatine Trial (FACT), we examined the independent and joint effects of FA and creatine treatment among adults recruited independent of folate status in Bangladesh, a country without FA fortification of food. The *a priori* primary outcome was change in total bAs concentrations at 12 wk. Among participants receiving 800 μg FA/d, there was a greater decrease in total bAs concentrations [percentage change in geometric mean (GM) of −17.8%] compared with PBO (−9.5%) ($p<0.05$).[17] Secondary outcomes included bAs metabolite concentrations (recently measured and reported here) and the proportions of As metabolites in urine (reported by Bozack et al.[18]). Among participants receiving 400 μg FA, 800 μg FA, or 400 μg FA + creatine daily, there were significantly greater mean decreases in urine %InAs and %MMAs [change in $\log(\%\mathrm{InAs}) = -0.09$, -0.14, and -0.11 for each treatment group, respectively; change in $\%\mathrm{MMAs} = -1.80$, -2.60, and -1.85, respectively] compared with PBO [changes in $\log(\%\mathrm{InAs})$ and $\%\mathrm{MMAs} = 0.05$ and 0.15, respectively] ($p<0.05$). In addition, participants receiving 400 μg FA, 800 μg FA, or 400 μg FA + creatine daily had a greater mean increase in %DMAs (3.25, 4.57, and 3.11, respectively) compared with PBO (−1.17) ($p<0.05$).[18] The decrease in urinary %MMAs among participants receiving 3 g creatine/d at week 1 (−0.90) and week 12 (−1.13) also exceeded that of the PBO group (change in % MMAs, week $1 = -0.62$) ($p<0.05$).[18]

Although the relative distribution of As metabolite proportions in urine is an informative indicator of As methylation capacity, the concentrations of As metabolites in blood may better reflect As exposure to tissues. In addition, as a biomarker of exposure, bAs is not complicated by adjustment for dilution, as urinary As is, or—as compared with water As—by difficulties associated with estimating As consumption through water or dietary sources.

The present study uses newly available bAs metabolite data from FACT to investigate the effect of FA and creatine supplementation on As metabolites in blood. We report the effects of supplementation on the bAs primary methylation index (PMI: MMAs/InAs) and the secondary methylation index (SMI: DMAs/MMAs), as well as metabolite concentrations: blood InAs (bInAs), bMMAs, and blood DMAs (bDMAs), in a population of Bangladeshi adults having a wide range of folate nutritional status.

Methods

Study Population and Design

FACT was conducted in Araihazar, Bangladesh, and has previously been described in detail.[17] FACT participants were recruited from the Health Effects of Arsenic Longitudinal Study (HEALS), a prospective cohort that originally comprised 11,746 adults living in a 25-km^2 area of Araihazar, Bangladesh, recruited in 2000.[19] HEALS was subsequently expanded to include >35,000 participants.[20] Married adults were eligible for enrollment in HEALS if they had resided in the study area for ≥5 y and had primarily obtained their drinking water from one of the study wells for ≥3 y. Between December 2009 and May 2011, a random subset of 622 participants were recruited from the HEALS cohort to participate in FACT. Five trained teams consisting of one interviewer and one physician recruited participants through house-to-house visits. Male and female participants were eligible for FACT if, upon medical examination, they were 20–65 years of age and had been drinking water from a household well with water As concentrations $>50\,\mu\mathrm{g/L}$ for ≥1 y. Medical conditions were determined by a study physician who conducted a thorough medical examination at the screening visit. This examination included assessment of current and past medications, physical exam, self-report, and review of medical records. Individuals were excluded if they were pregnant, taking nutritional supplements, had protein in their urine, renal diseases, diabetes, or other health problems, such as gastrointestinal issues.

FACT participants were randomly assigned to one of the following treatment groups: *a*) PBO ($n=104$), *b*) 400 μg FA/d (referred to hereafter as 400FA; $n=156$), *c*) 800 μg FA/d (800FA; $n=154$), *d*) 3 g creatine/d (creatine; $n=104$), and *e*) 3 g creatine and 400 μg FA/d (creatine + 400FA; $n=104$). The dose of 400 μg FA/d was selected because it is the U.S. recommended dietary allowance (RDA)[21]; the dose of 800 μg FA/d was selected to

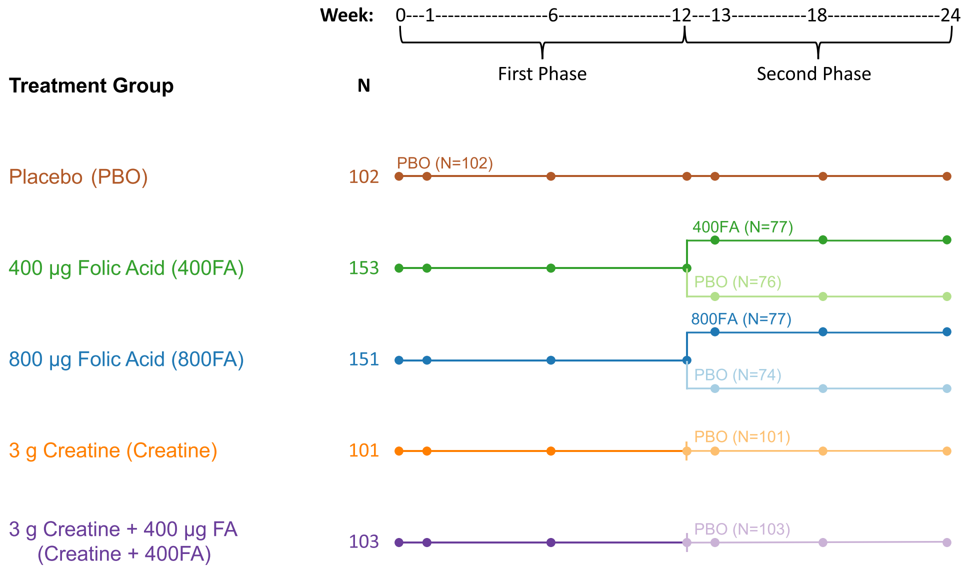

Figure 2. Folic Acid and Creatine Trial (FACT) study design overview. The PBO, 400 µg of FA/d (400FA), 800 µg of FA/d (800FA), 3 g creatine (creatine), and 3 g creatine + 400 µg of FA/d (creatine + 400FA) are shown in brown, green, blue, orange, and purple, respectively. All five treatments were administered daily, with participants switching to PBO at 12 wk as indicated with a lighter color. Modified from Peters et al.[17] Note: FA, folic acid; PBO, placebo.

evaluate whether a higher dose would be more effective; and the dose of 3 g creatine/d was selected to exceed the average daily creatine loss for a 70-kg male individual (~ 1.9 g/d) as estimated by Brosnan et al., that is, it was a dose predicted to down-regulate endogenous creatine biosynthesis.[14] To reduce bias and ensure a balance of male and female participants in each treatment arm, male and female participants were randomized separately in blocks. By blocks, treatment groups were assigned in the ratio of 1 (PBO):1.5 (400FA):1.5 (800FA):1 (creatine):1 (creatine + 400FA). Random permutation was used to assign treatment groups. With the exception of the data management specialists who assigned treatment groups, all participants, field staff, village health workers, laboratory technicians, and investigators were blinded to the treatment for the duration of the study.

The study included two phases (Figure 2). In the first 12-wk phase, participants received supplementation according to treatment group assignments, and in the second 12-wk phase, half of the participants in the 400FA and 800FA groups were randomly switched to PBO (400FA/PBO: $n=76$ and 800FA/PBO: $n=74$) and half continued FA supplementation (400FA/FA: $n=77$; 800FA/FA: $n=77$) to assess the potential reversal of treatment effects on As metabolites following cessation of FA supplementation. During the second phase, all participants in the PBO, creatine, and creatine + 400FA groups received PBO to maintain the study blind.

Ethics

The FACT protocol was approved by the institutional review board at Columbia University and the Bangladesh Medical Research Council. Informed consent was obtained by staff physicians in Bangladesh.

Fieldwork and Study Follow-Up

Fieldwork was conducted by the five trained teams, each comprising one interviewer and one physician. Sociodemographic characteristics were obtained at enrollment by field staff–administered questionnaires designed specifically for this study. During home visits, blood samples were collected at baseline and at weeks 1, 12, and 24.

At the beginning of the trial, participants received READ-F As-removal water filters (Brota Services International) and were encouraged to use the filters for all drinking and cooking water. Efficacy of the filters was tested in the field during home visits using the Hach EZ kit (Hach Company); filters were repaired or replaced if As water concentrations exceeded 10 µg/L or if participants reported failure of the filter[22] (~ 50 of the 622 filters failed during the trial). Arsenic concentrations were also measured by laboratory analysis in a random subset of filtered water collected at baseline and at weeks 12 and 24 (described below). Total urinary As concentrations throughout the study were used as a proxy of water filter use. In addition, water filter use 1 y after FACT ended was measured by a follow-up survey, as previously reported by Sanchez et al.[22]

In the first phase of the trial, all participants received two pill bottles containing *a*) FA or matched PBO pills and *b*) creatine or matched PBO pills. In the second phase, all participants received one pill bottle containing FA or matched PBO pills. Village health workers performed daily home visits to ask about compliance or observe participants taking pills. Compliance was assessed using pill counts and circulating folate concentrations as follows: *a*) the percentage of study pills taken, and *b*) changes in red blood cell and plasma folate concentrations as reported in changes in GMs over time.[17]

Sample Handling and Laboratory Measures

Sample handling and laboratory methods have been described in detail.[17] Briefly, venous blood was collected in vacutainer tubes with ethylenediaminetetraacetic acid during field visits, stored in IsoRack cool packs (Brinkmann Instruments) at 4°C, and transported to our Araihazar field clinic within 4 h. Plasma was separated by centrifugation at $1{,}350\times g$ for 10 min at 25°C. Whole blood and plasma samples were stored at −80°C before shipment to Columbia University on dry ice.

Total bAs was measured using a PerkinElmer Elan DRC II inductively coupled plasma mass spectrometer (ICP-MS) with an AS 93 + autosampler [intra- and interassay coefficient of variations (CVs): 2.7% and 5.7%, respectively]. As^{III}, arsenate (As^{V}), MMAs, and DMAs were measured in blood and urine using high-performance liquid chromatography (HPLC) coupled to dynamic reaction cell ICP-MS, as previously described.[23,24] This method cannot differentiate between reduced and oxidized forms of MMAs and DMAs. Although this method can separate As^{V} from As^{III}, samples can oxidize during sample processing; therefore, As^{V} and As^{III} were added and reported as a single variable reflecting total InAs. The limit of detection (LOD) and interassay CVs were, respectively, 0.11 μg/L and 15.9% for arsenobetaine + arsenocholine, 0.22 μg/L and 4.4% for As^{III} and As^{V}, 0.22 μg/L and 4.4% for MMAs, and 0.22 μg/L and 4.7% for DMAs.

Plasma folate and vitamin B_{12} were analyzed using a radio protein-binding assay according to the manufacturer's instructions (SimulTRAC-SNB; MP Biomedicals). Intra- and interassay CVs for plasma folate were 5% and 13%, respectively, and for vitamin B_{12} were 6% and 17%, respectively. Plasma total homocysteine (tHcy) concentrations were measured by HPLC with fluorescence detection according to method described by Pfeiffer et al.[25] Separation was achieved with a 150 × 3.2-mm, 5-μm Prodigy ODS2 analytic column (Phenomenex). Fluorescence was detected with a Waters 474 fluorescence detector (Waters Corporation) with excitation at 385 nm and emission at 515 nm. L-Homocysteine and L-cysteine were used as external calibrators.[25] The intra- and interassay CVs for tHcy were 5% and 7%, respectively.

Baseline water As concentrations were measured in nonfiltered water samples collected in polyethylene scintillation vials. Samples were acidified to 1% using high-purity Optima HCl (Fisher Scientific) for 48 h, diluted 1:10, and analyzed with high-resolution ICP-MS.[26] No samples were below the LOD of <0.2 μg/L, and the intra- and interassay CVs were 2% and 2.6%, respectively.

Study Sample

A total of 622 participants were recruited. Six participants discontinued the trial owing to adverse events [PBO group (1 participant): abdominal cramps; 400FA group (1 participant): hypertension; 800FA group (3 participants): abdominal cramps, severe vertigo, bilateral hydronephrosis; creatine group (1 participant): severe vertigo] and 3 participants discontinued owing to pregnancy (400FA, creatine, and creatine + 400FA groups). Two participants dropped out of the trial (PBO and 400FA groups), and 1 participant was excluded from this analysis owing to a missing blood sample. A total of 609 participants completed the study and were used for the present analyses (PBO: $n = 102$, 400FA: $n = 153$, 800FA: $n = 150$, creatine: $n = 101$, creatine + 400FA: $n = 103$).

Statistical Methods

Primary and secondary bAs methylation indices were calculated as bMMAs/bInAs concentrations and bDMAs/bMMAs concentrations, respectively. Baseline summary statistics were calculated for participant sociodemographic data, baseline As methylation indices and metabolites, and nutritional parameters using GMs and geometric standard deviations (GSDs) for continuous variables and frequencies for categorical variables. GMs are commonly used in analyses involving environmental chemicals and toxicants, such as metals or metalloids that are measured in blood and urine, because they provide a more accurate measure of the central tendency of lognormally distributed variables, such as bAs metabolites, as compared with arithmetic means.[27] Differences between treatment groups at baseline were detected using the Kruskall–Wallis test for continuous variables and chi-square test for categorical variables. Because the distributions of bAs methylation indices and metabolites were right skewed, these variables were natural log-transformed to meet model assumptions, reduce the impact of extreme values, and improve model fitting.

Linear models with repeated measures were used to examine treatment effects on bAs methylation indices and metabolite concentrations in the first 12-wk phase and the potential reversal of treatment effects on As metabolites following FA cessation during phase 2 (weeks 12–24). Model parameters were estimated using generalized estimating equations with an exchangeable correlation structure to account for within-person correlation in repeated measures over time; robust standard errors were used for statistical inference. The models for phase 1 of the RCT included the variables of baseline total bAs, treatment group for the five arms, time (baseline, week 1, and week 12), and group-by-time interactions. Coefficients of interaction terms indicate differences between each treatment group and the PBO group in the mean within-person change from baseline. A Wald test on the coefficients that define group-by-time interactions was used to detect differences in the mean within-person change between specific treatment groups. In addition, to explore potential sex-differences, models examining the treatment effects on bAs methylation indices and metabolite concentrations during phase 1 were further stratified by sex. Effect modification was also assessed using models including group-by-time-by-sex interaction terms. The models for phase 2 of the RCT (effect of FA cessation) included variables for baseline total bAs, time (baseline, week 12, and week 24), treatment groups (two FA doses over 24 wk and two FA doses switched to PBO in the second 12 wk), and interactions of group-by-time. Percentage change in GM over a specific time interval for each treatment group was derived from the estimated model parameters with a 95% CI. All analyses were performed using R (version 3.6.3; R Development Core Team).

Results

By design, approximately half of participants were male (50.2%, $n = 306$). The age (GM ± GSD) of participants ranged from 24 to 55 y (37.5 ± 1.2 y). The prevalence of folate deficiency (<9 nmol/L in plasma) was 19.7% ($n = 120$) in this study population. In addition, 38.8% ($n = 236$) of participants had hyperhomocysteinemia (plasma homocysteine ≥13 μmol/L) and 24% ($n = 146$) were B_{12} deficient (<151 pmol/L in plasma). GM baseline well water As concentrations across treatment groups ranged from 119 to 128 μg/L, that is, they were ∼12–15 times the WHO-recommended limit of 10 μg/L[3] but were not statistically different across treatment groups ($p = 0.95$). There were no significant differences in participant characteristics between treatment groups at baseline ($p > 0.05$) (Table 1).

Compliance with supplementation in FACT has been previously reported by Peters et al.[17] Pill counts were used to calculate the percentage of study pills taken by each participant throughout the trial. Median compliance was 99.5%, and compliance was similar across treatment groups and study phases (range: 79.1%–100%; IQR: 98.3%–100.0%).[17] Compliance, as assessed by plasma folate, was also high. Plasma folate concentrations at week 12 in the 400FA, 800FA, and creatine + 400FA groups were significantly greater than baseline concentrations (paired Wilcoxon test $p < 0.001$) and significantly greater than week 12 concentrations in the PBO group (Wilcoxon test $p < 0.001$). The prevalence of folate deficiency did not considerably change in the PBO and creatine arms, whereas all groups receiving FA had a decrease in the prevalence of folate deficiency to <1.5% at week 12 (from 17.9%–23.5% at baseline). Furthermore, participants who remained on FA for the second phase of the trial sustained their plasma folate concentrations, whereas plasma folate returned to near baseline levels among participants who switched to PBO for the second phase.[17]

Table 1. Folic Acid and Creatine Trial study participant characteristics [GM ± GSD or *n* (%)] at baseline by treatment group.

Characteristic	PBO ($n=102$)	400FA ($n=153$)	800FA ($n=150$)	Creatine ($n=101$)	Creatine + 400FA ($n=103$)	*p*-Value[a]
Age (y)	37.3 ± 1.2	38.2 ± 1.2	37.3 ± 1.3	37.4 ± 1.3	37.2 ± 1.2	0.85
Sex (*n*)						>0.99
Female	51 (50.0)	76 (49.7)	75 (50.0)	50 (49.5)	51 (49.5)	
Male	51 (50.0)	77 (50.3)	75 (50.0)	51 (50.5)	52 (50.5)	
Smoking status (*n*)[b]						0.69
Never	77 (75.5)	115 (75.2)	106 (70.7)	72 (71.3)	72 (69.9)	
Ever	25 (24.5)	36 (23.5)	44 (29.3)	29 (28.7)	31 (30.1)	
Betel nut use (*n*)[b]						0.76
Never	73 (71.6)	115 (75.2)	113 (75.3)	76 (75.2)	82 (79.6)	
Ever	29 (28.4)	36 (23.5)	37 (24.7)	25 (24.8)	21 (20.4)	
Land ownership (*n*)[c]						0.85
Does not own land	55 (53.9)	76 (49.7)	78 (52.0)	53 (52.5)	58 (56.3)	
Owns land	47 (46.1)	77 (50.3)	72 (48.0)	48 (47.5)	44 (42.7)	
Body mass index (kg/m^2)[d]	20.2 ± 1.2	19.4 ± 1.1	19.7 ± 1.1	19.8 ± 1.2	19.3 ± 1.1	0.30
Water arsenic (μg/L)	123 ± 2	119 ± 2	119 ± 2	125 ± 2	128 ± 2	0.95
Urinary arsenic (μg/L)	86.2 ± 2.4	98.6 ± 2.5	97.2 ± 2.4	110.0 ± 2.5	117.0 ± 2.3	0.38
Blood arsenic[e]	8.03 ± 1.80	8.26 ± 1.88	8.37 ± 1.72	8.24 ± 1.85	8.76 ± 1.68	0.64
PMI[e]	1.61 ± 1.27	1.65 ± 1.20	1.63 ± 1.20	1.63 ± 1.20	1.70 ± 1.19	0.56
SMI[e]	0.62 ± 1.29	0.63 ± 1.29	0.65 ± 1.31	0.63 ± 1.29	0.66 ± 1.30	0.29
InAs (μg/L)[e]	2.20 ± 1.75	2.22 ± 1.79	2.25 ± 1.65	2.24 ± 1.85	2.27 ± 1.65	0.86
MMAs (μg/L)[e]	3.55 ± 1.89	3.67 ± 1.97	3.67 ± 1.82	3.64 ± 1.93	3.85 ± 1.73	0.79
DMAs (μg/L)[e]	2.19 ± 1.83	2.30 ± 1.93	2.38 ± 1.75	2.29 ± 1.83	2.56 ± 1.73	0.37
%InAs[e]	27.5 ± 1.2	26.9 ± 1.2	26.9 ± 1.1	27.1 ± 1.1	25.9 ± 1.1	0.05
%MMAs[e]	44.2 ± 1.1	44.3 ± 1.1	43.8 ± 1.1	44.2 ± 1.1	44.0 ± 1.1	0.75
%DMAs[e]	27.2 ± 1.2	27.8 ± 1.2	28.4 ± 1.2	27.8 ± 1.2	29.2 ± 1.2	0.08
Urinary creatinine (mg/dL)	38.6 ± 2.1	43.7 ± 2.2	40.9 ± 2.2	46.2 ± 2.0	47.5 ± 2.2	0.35
Red blood cell folate (nmol/L)[f]	451 ± 1	448 ± 2	468 ± 1	NA	NA	0.63
Plasma folate (nmol/L)	13.6 ± 1.8	13.5 ± 1.9	14.5 ± 1.8	14.4 ± 1.6	13.3 ± 1.7	0.66
Folate deficient (<9 nmol/L in plasma) (*n*)	22 (21.6)	36 (23.5)	27 (18.0)	14 (13.9)	21 (20.4)	0.39
Plasma homocysteine (μmol/L)	12.2 ± 1.6	11.9 ± 1.6	11.9 ± 1.6	11.4 ± 1.5	11.8 ± 1.5	0.90
Hyperhomocysteinemia (≥13 μmol/L) (*n*)	43 (42.2)	56 (36.6)	59 (39.3)	39 (38.6)	39 (37.9)	0.93
Plasma vitamin B_{12} (pmol/L)	204 ± 2	216 ± 2	214 ± 2	224 ± 2	210 ± 2	0.82
Vitamin B_{12} deficient (<151 pmol/L) (*n*)	25 (24.5)	37 (24.2)	39 (26.0)	20 (19.8)	25 (24.3)	0.86

Note: %, percentage; 400FA, 400 μg FA/d; 800FA, 800 μg FA/d; creatine, 3 g creatine/d; creatine + 400FA, 3 g creatine and 400 μg FA/d; DMAs, dimethyl-arsenical species; FA, folic acid; GM, geometric mean; GSD, geometric standard deviation; InAs, inorganic arsenic; MMAs, monomethyl-arsenical species; NA, not available/not measured for creatine and creatine + 400 FA groups; PBO, placebo; PMI, primary methylation index; SMI, secondary methylation index.

[a]Kruskal–Wallis test for continuous variables and chi-square test for categorical variables.

[b]400FA: $n=151$.

[c]Creatine + 400FA: $n=102$.

[d]PBO: $n=101$; 400FA: $n=150$; 800FA: $n=147$; creatine: $n=98$; creatine + 400FA: $n=102$.

[e]400FA: $n=152$; creatine: $n=100$.

[f]PBO: $n=100$; 400FA: $n=148$; 800FA: $n=148$.

Treatment Effects on bAs Methylation Indices

GMs of PMI and SMI at baseline, week 1, and week 12 are provided in Table S1; individual-level data are provided in Excel File 1. Changes in bAs indices between baseline and week 12, ranked by change in PMI and SMI, are shown in Figure S1. The percentage change in the GM of bAs methylation indices between baseline and week 12 are shown in Table 2. As early as week 1, the increase in SMI, an indicator of the extent to which the second methylation reaction is completed, was higher in the creatine + 400FA treatment group than the PBO group: within-person change increased by 6.3% (95% CI: 3.4, 9.3) compared with 2.1% (95% CI: −0.8, 5.2) for PBO ($p=0.05$).

At week 1, the increase in SMI from baseline was greater among participants in the 400FA, 800FA, and creatine treatment arms, with respective within-person changes of 5.4% (95% CI: 2.4, 8.5), 3.0% (95% CI: 0.2, 6.3), and 3.9% (95% CI: 1.4, 6.5) compared with PBO, but it did not reach statistical significance ($p=0.12$, 0.69, and 0.37, respectively). For PMI, in the 400FA, creatine, and creatine + 400FA treatment groups, the −18.1% (95% CI: −21.1, −15.0), −18.3% (95% CI: −22.2, −14.2), and −20.4% (95% CI: −23.5, −17.2) decreases at week 1, respectively, were greater than the −17.0% (95% CI: −21.6, −12.0) decrease in PBO but did not reach statistical significance ($p=0.69$, 0.66, and 0.23, respectively). In the 800FA group, the decrease in PMI of −15.3% (95% CI: −18.3, −12.1) was less than that of PBO, although it did not reach statistical significance ($p=0.57$).

At week 12, the mean within-person decreases from baseline in PMI among participants in the 400FA, 800FA, and creatine + FA treatment groups were −12.6% (95% CI: −15.7, −9.4), −10.8% (95% CI: −13.9, −7.7), and −12.7% (95% CI: −16.1, −9.2), respectively; these were all significantly greater than the −5.5% (95% CI: −9.8, −1.0) decrease in the PBO group. Among participants in the creatine group, the mean within-person decrease in PMI of −9.8% (95% CI: −14.9, −4.5) was larger than the PBO group, but this did not reach statistical significance ($p=0.22$). In addition, among all treatment groups, the mean percentage changes in SMI at week 12 were significantly greater ($p<0.05$) than the 2.2% (95% CI: −1.5, 5.9) increase in the PBO group: 400FA: 26.3% (95% CI: 22.1, 30.5), 800FA: 23.8% (95% CI: 19.7, 27.9), creatine: 8.9% (95% CI: 4.7, 13.2), and creatine + 400FA: 17.9% (95% CI: 14.3, 21.6).

Treatment Effects on bAs Metabolite Concentrations

GMs of bInAs, bMMAs, and bDMAs concentrations at baseline, week 1, and week 12 are provided in Table S1; individual-level data are provided in Excel File 1. The changes in participants'

Table 2. GM ± GSD at baseline and percentage change in GM (95% CI) at interval (week) of blood arsenic methylation indices and metabolite concentrations from baseline by treatment group.

	PBO	400FA		800FA		Creatine		Creatine + 400FA	
Arsenic methylation measures by timepoint (week)	GM ± GSD or % change from baseline (95% CI)	GM ± GSD or % change from baseline (95% CI)	p-Value[a]	GM ± GSD or % change from baseline (95% CI)	p-Value[a]	GM ± GSD or % change from baseline (95% CI)	p-Value[a]	GM ± GSD or % change from baseline (95% CI)	p-Value[a]
Arsenic methylation indices									
PMI									
Baseline	1.61 ± 1.27	1.65 ± 1.20	—	1.63 ± 1.20	—	1.63 ± 1.20	—	1.70 ± 1.19	—
1	−17.0 (−21.6, −12.0)	−18.1 (−21.1, −15.0)	0.69	−15.3 (−18.3, −12.1)	0.57	−18.3 (−22.2, −14.2)	0.66	−20.4 (−23.5, −17.2)	0.23
12	−5.5 (−9.8, −1.0)	−12.6 (−15.7, −9.4)	0.01	−10.8 (−13.9, −7.7)	0.05	−9.8 (−14.9, −4.5)	0.22	−12.7 (−16.1, −9.2)	0.01
SMI									
Baseline	0.62 ± 1.29	0.63 ± 1.29	—	0.65 ± 1.31	—	0.63 ± 1.29	—	0.66 ± 1.30	—
1	2.1 (−0.8, 5.2)	5.4 (2.4, 8.5)	0.12	3.0 (0.2, 6.3)	0.69	3.9 (1.4, 6.5)	0.37	6.3 (3.4, 9.3)	0.05
12	2.2 (−1.5, 5.9)	26.3 (22.1, 30.5)	<0.001	23.8 (19.7, 27.9)	<0.001	8.9 (4.7, 13.2)	0.02	17.9 (14.3, 21.6)	<0.001
Arsenic metabolites (μg/L)									
InAs									
Baseline	2.20 ± 1.75	2.22 ± 1.79	—	2.25 ± 1.65	—	2.24 ± 1.85	—	2.27 ± 1.65	—
1	13.5 (9.7, 17.5)	13.7 (10.9, 16.5)	0.96	11.6 (8.9, 14.4)	0.42	14.3 (10.5, 18.1)	0.80	15.9 (12.7, 19.1)	0.36
12	4.0 (0.9, 7.2)	2.9 (0.5, 5.5)	0.61	2.1 (0.3, 4.6)	0.36	4.8 (1.5, 8.3)	0.73	5.4 (2.6, 8.3)	0.51
MMAs									
Baseline	3.55 ± 1.89	3.67 ± 1.97	—	3.67 ± 1.82	—	3.64 ± 1.93	—	3.85 ± 1.73	—
1	−6.1 (−8.4, −3.7)	−7.2 (−8.7, −5.7)	0.42	−5.9 (−7.4, −4.4)	0.92	−7.2 (8.8, 5.5)	0.45	−8.2 (−9.6, −6.8)	0.11
12	−2.0 (−4.0, 0.0)	−10.3 (−11.9, −8.8)	<0.001	−9.5 (−11.1, −8.0)	<0.001	−5.9 (−8.6, −3.0)	0.03	−8.4 (−10.0, −6.9)	<0.001
DMAs									
Baseline	2.19 ± 1.83	2.30 ± 1.93	—	2.38 ± 1.75	—	2.29 ± 1.83	—	2.56 ± 1.73	—
1	−4.4 (−6.5, −2.2)	−2.4 (−4.4, −0.4)	0.20	−3.5 (−5.7, −1.2)	0.57	−3.9 (−5.8, −1.9)	0.74	−2.8 (−4.8, −0.8)	0.30
12	−0.1 (−2.8, 2.6)	12.8 (10.5, 15.2)	<0.001	11.3 (8.9, 13.8)	<0.001	2.2 (0.5, 5.0)	0.23	7.5 (5.2, 9.7)	<0.001

Note: Percentage change was estimated using linear models. GMs for each time point are included in Table S1; individual-level data are included in Excel File 1. —, not applicable; %, percentage; 400FA, 400 μg FA/d; 800FA, 800 μg FA/d; CI, confidence interval; creatine, 3 g creatine/d; creatine + 400FA, 3 g creatine and 400 μg FA/d; DMAs, dimethyl-arsenical species; FA, folic acid; GM, geometric mean; GSD, geometric standard deviation; InAs, inorganic arsenic; MMAs, monomethyl-arsenical species; PBO, placebo; PMI, primary methylation index; SMI, secondary methylation index.

[a]p-Values were derived from Wald test on coefficients of group-by-time interaction terms in the linear models with repeated measures for differences in changes from baseline between each treatment and PBO group.

bAs metabolite concentrations between baseline and week 12 by treatment group are shown in Figure 3. Decreases in bAs metabolite concentrations between baseline and week 12 were observed across treatment groups, likely due to water filter use.[22] However, Figure 3 illustrates that all treatment groups who received FA showed an overall trend toward a decrease in InAs and MMAs concentrations between baseline and week 12 (i.e., a greater proportion of participants falling below the *y*-axis, representing a decrease in concentrations).

The percentage change in the GM of bAs metabolite concentrations over time in phase 1 are shown in Table 2. At week 1, compared with baseline, bMMAs decreased across all treatment groups, although these changes were not significantly different than that of the PBO group ($p > 0.05$). At week 12, the decreases in bMMAs were significantly greater ($p < 0.05$) for all treatment groups receiving FA as compared with PBO: the mean within-person decrease was −2.0% (95% CI: −4.0, 0.0) in the PBO group, vs. −10.3% (95% CI: −11.9, −8.8), −9.5% (95% CI: −11.1, −8.0), and −8.4% (95% CI: −10.0, −6.9) for the 400FA, 800FA, and creatine + 400FA groups, respectively. The mean within-person decrease in bMMAs in the 800FA group (−9.5%) was not significantly different from the decrease in the 400FA group (−10.3%) (Wald test $p > 0.05$), suggesting that there was not a dose–response effect of FA on bMMAs. The mean within-person change in bDMAs concentrations from baseline to week 12 was 12.8% (95% CI: 10.5, 15.2) for the 400FA group and 11.3% (95% CI: 8.9, 13.8) for the 800FA group vs. −0.1% (95% CI: −2.8, −2.6) for the PBO group ($p < 0.001$). The changes in bInAs concentrations from baseline to weeks 1 and 12 in the FA and creatine treatment groups were not significantly different from changes in the PBO group ($p > 0.05$).

Overall, models stratified by sex were consistent with the unstratified models. Results from sex-stratified linear models are presented in Tables S2 and S3; GMs of bAs methylation indices and metabolite concentrations are provided in Tables S4 and S5. There appear to be sex-specific changes in bAs indices and concentrations that were significantly different ($p < 0.05$) in the creatine group compared with PBO among male participants that were not significant ($p > 0.05$) among female participants. This was reflected in decreases in bMMAs and increases in SMI in the creatine arm that were significantly different from PBO in male participants [creatine vs. PBO: bMMAs: −6.6% (95% CI: −9.5, −3.7) vs. −2.7% (95% CI: −4.8, −0.5), $p = 0.03$; SMI: 12.3% (95% CI: 6.1, 18.8) vs. 3.1% (95% CI: −0.9, 7.3), $p = 0.02$], but not female participants [bMMAs: −5.0% (95% CI: −9.6, −0.2) vs. −1.4% (95% CI: −4.7, 2.1), $p = 0.31$; SMI: 5.4% (95% CI: 0.2, 10.9) vs. 1.3% (95% CI: −4.7, 7.6), $p = 0.21$]. However, effect modification of creatine treatment by sex was not statistically significant as measured by the inclusion of a group-by-time-by-sex interaction term in unstratified models ($p > 0.05$; Table S6).

Treatment Effects during the Second Phase

To evaluate the potential reversal of treatment effects on As metabolites after cessation of FA, we evaluated the percentage changes in bAs indices and metabolites from baseline to week 12, baseline to week 24, and from week 12 to week 24 (Table 3). The percentage increase in SMI between baseline and week 24 was significantly greater among the groups who continued FA treatment compared with groups who discontinued treatment [400FA/FA: 29.2% (95% CI: 23.1, 35.5), 400FA/PBO: 14.1% (95% CI: 9.13, 19.3), 800FA/FA: 23.7% (95% CI: 18.9, 28.6), 800FA/PBO: 13.4% (95% CI: 8.2, 18.9)] ($p < 0.05$). In the 800FA/FA group, percentage changes in bInAs, bMMAs, bDMAs, and PMI over 24 wk were similar to those in the 800FA/PBO group ($p > 0.05$). Data from the second phase, comparing week 12 to week 24, showed

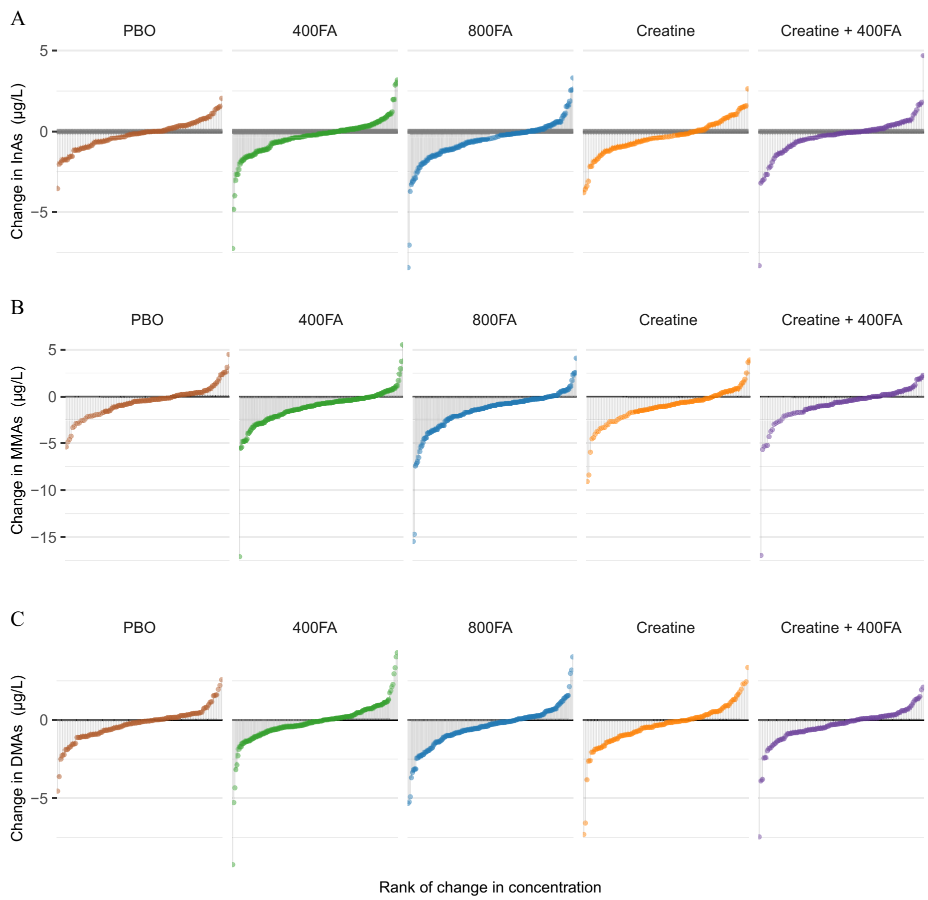

Figure 3. Change in participants' blood arsenic metabolite concentrations between baseline and 12 wk by treatment group. The *x*-axes are ranked by participants with the largest decreases to the largest increases in blood arsenic metabolite concentrations. The PBO, 400 µg of FA/d (400FA), 800 µg of FA/d (800FA), 3 g creatine (creatine), and 3 g creatine + 400 µg of FA/d (creatine + 400FA) are shown in brown, green, blue, orange, and purple, respectively. Individual-level blood arsenic metabolite concentration data are included in Excel File 1. Note: DMAs, dimethyl-arsenical species; FA, folic acid; InAs, inorganic arsenic; MMAs, monomethyl-arsenical species; PBO, placebo.

Table 3. Percentage change (95% CI) in blood arsenic methylation indices and metabolite concentrations between baseline and week 12, baseline and week 24, and week 12 and week 24 by folic acid supplementation groups.

Arsenic methylation measures by treatment group	400FA[a]			800FA[b]		
	Baseline to week 12	Baseline to week 24	Week 12 to week 24	Baseline to week 12	Baseline to week 24	Week 12 to week 24
Arsenic methylation indices						
PMI						
FA/FA	−11.2 (−15.3, −6.8)	−11.1 (−14.8, −7.2)	−5.4 (−13.3, 2.0)	−8.1 (−12.1, −4.0)	−9.5 (−13.3, −5.6)	−7.2 (−14.3, −0.5)
FA/PBO	−13.3 (−18.0, −8.4)	−6.4 (−10.6, −2.0)	2.3 (−4.9, 9.0)	−12.4 (−17.1, −7.5)	−9.2 (−13.0, −5.3)	−1.8 (−10.0, 5.7)
SMI						
FA/FA	26.9 (21.1, 33.1)	29.2 (23.1, 35.5)	3.7 (−2.1, 9.1)	20.9 (16.0, 26.0)	23.7 (19.0, 28.6)	4.2 (−1.0, 9.1)
FA/PBO	25.2 (19.6, 31.1)	14.1 (9.1, 19.3)	−7.6 (−14.5, −1.1)	26.1 (19.8, 32.7)	13.4 (8.2, 18.9)	−9.0 (−14.8, −3.5)
Arsenic metabolites						
InAs						
FA/FA	1.6 (−1.5, −4.8)	0.9 (−2.2, 4.0)	3.0 (−1.5, 7.2)	0.7 (−2.3, 3.7)	0.9 (−2.0, 3.9)	3.9 (−0.2, 7.2)
FA/PBO	3.6 (−0.1, 7.6)	1.0 (−2.1, 4.2)	1.1 (−3.6, 5.5)	2.4 (−1.4, 6.4)	3.4 (0.2, 6.8)	4.5 (−0.7, 9.5)
MMAs						
FA/FA	−10.0 (−12.2, −7.8)	−10.4 (−12.3, −8.5)	−2.4 (−5.9, 1.1)	−7.8 (−9.6, −5.9)	−8.8 (−10.5, −7.1)	−3.1 (−6.2, −0.1)
FA/PBO	−10.3 (−12.6, 8.0)	−5.5 (−7.6, −3.4)	3.3 (−0.1, 6.6)	−10.7 (−13.1, −8.2)	−6.3 (−8.3, −4.3)	2.82 (−0.3, 5.8)
DMAs						
FA/FA	14.1 (10.8, 17.5)	15.6 (11.8, 19.6)	1.3 (−2.5, 5.0)	11.4 (8.2, 14.6)	6.1 (2.7, 9.6)	1.2 (−2.4, 4.6)
FA/PBO	12.2 (9.0, 15.5)	7.8 (4.7, 10.9)	−4.1 (−8.4, 0.1)	12.4 (−8.8, 16.1)	6.1 (2.7, 9.6)	−5.9 (−10.2, −1.8)

Note: Percentage change was estimated using linear models. GMs for each time point are included in Table S7; individual-level data are included in Excel File 1. 400FA, 400 µg FA/d; 800FA, 800 µg FA/d; CI, confidence interval; DMAs, dimethyl-arsenical species; FA, folic acid; FA/FA, continued folic acid treatment; FA/PBO, discontinued folic acid treatment; InAs, inorganic arsenic; MMAs, monomethyl-arsenical species; PBO, placebo; PMI, primary methylation index; SMI, secondary methylation index.

[a]400FA/FA: $n=77$; 400FA/PBO: $n=76$.

[b]800FA/FA: $n=76$; 800FA/PBO: $n=74$.

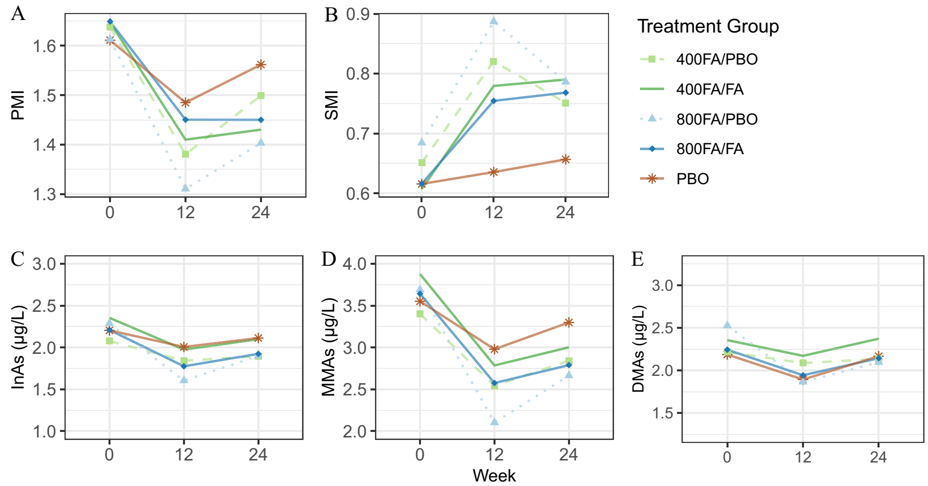

Figure 4. Geometric mean of blood arsenic indices and metabolite concentrations by treatment group from baseline to week 24. The *y*-axes are the (A) primary methylation index (PMI), (B) secondary methylation index (SMI), (C) inorganic arsenic (InAs), (D) monomethyl-arsenical species (MMAs), and (E) dimethyl-arsenical species (DMAs) levels and the *x*-axes are weeks. Participants that received PBO, continuous 400 μg of FA/d (400FA/FA), continuous 800 μg of FA/d (800FA/FA), cessation of 400 μg FA/d at 12 wk (400FA/PBO), or cessation of 800 μg of FA/d at 12 wk (800FA/PBO) are shown in brown solid lines (triangle points), green solid lines (circle points), blue solid lines (square points), light green dashed lines (circle points), and light blue dotted lines (square points), respectively. The bAs PMI and SMI were calculated as bMMAs/bInAs and bDMAs/bMMAs concentrations, respectively. Percentage change was estimated using linear models. Individual-level data are included in Excel File 1. Note: bDMAs, blood dimethyl-arsenical species; bInAs, blood inorganic arsenic metabolites; bMMAs, blood monomethyl-arsenical species; FA, folic acid; PBO, placebo.

evidence of the reversal of treatment effects on As metabolites following FA cessation, with significant decreases in SMI [−9.0% (95% CI: −14.8, −3.5)] and DMAs [−5.9% (95% CI: −10.2, −1.8)] in the 800FA/PBO group and similar trends in the 400FA/PBO group that were not statistically significant ($p > 0.05$). For those who continued 800 μg FA supplementation, PMI and MMAs concentrations decreased between weeks 12 and 24 [PMI: −7.2% (95% CI: −14.3, −0.5); MMAs: −3.1% (95% CI: −6.2, −0.1)], whereas changes in the 800FA/PBO group were not significantly different than 0 [PMI: −1.8% (95% CI: −10.0, 5.7), MMAs: 2.8% (95% CI: −0.3, 5.8)]. Similar trends were observed in the 400FA/FA group but were not statistically significant ($p > 0.05$).

In Figure 4 and Table S7, the GMs for As methylation indices and bAs metabolite concentrations are shown at baseline, week 12, and week 24 for the PBO, 400FA, and 800FA treated groups, with separate groups for those who continued and discontinued FA supplementation; individual-level data are provided in Excel File 1. Across treatment groups, there was a trend toward an increase in bAs metabolite concentrations during the second phase, a period with reduced compliance with water filter use, which has previously been reported by Sanchez et al.[22] The plots illustrate similar declines in PMI and increases in SMI for the 400FA and 800FA groups at week 12; these were maintained at week 24 in both FA/FA groups, whereas in the FA/PBO groups, week 24 values tended to revert toward baseline values.

Discussion

Reduced As methylation capacity has been linked to increased risk for a number of As-related health outcomes, including cancers of the bladder,[28,29] lung,[12,30] and skin,[31,32] and noncancer outcomes, such as arsenical skin lesions[33,34] and atherosclerosis.[35] FACT was designed to investigate the efficacy of FA, creatine, and a combination of FA and creatine supplementation on increasing As methylation capacity among a population of mixed folate-deficient and -sufficient individuals in Bangladesh, a country that does not currently have food folate fortification. Our outcomes of interest included changes in bAs methylation indices, PMI and SMI, and bAs metabolite concentrations.

We previously reported findings on total bAs and urinary As metabolites.[8,17,18] In short, groups receiving FA supplementation had significantly greater decreases in %MMAs and increases %DMAs in urine compared with PBO ($p < 0.05$).[18] In our earlier RCT of 400 μg FA supplementation in only folate-deficient individuals,[8] the group receiving 400 μg FA had a significantly greater decrease in total bAs compared with PBO ($p < 0.05$). In contrast, in the present study of a mixed folate-deficient/replete population, 400 μg FA was not effective in lowering total bAs to a significantly greater extent than PBO over the course of the trial, indicating that a higher dose of FA (800 μg FA) or a longer duration of supplementation may be required to lower total bAs.[17]

Here we report new data on As metabolite concentrations in blood, where the distribution of As metabolites differs considerably from that previously reported for urine.[18] Although the distribution of As metabolites in urine at baseline ranged from 13.2% to 14.7% InAs, 12.5% to 13.4% MMAs, and 72.0% to 74.4% DMAs across treatment groups,[18] the GMs in blood ranged from 25.9% to 27.5% InAs, 44.0% to 44.3% MMAs, and 27.2% to 29.2% DMAs (Table 1). These differences between blood and urine are similar to those observed in our earlier study of folate-deficient participants[8] and are related to the more rapid renal excretion of DMAs in urine. The measurement of As, As indices, and As metabolite concentrations in blood has the added advantage that the overall concentration of blood is relatively constant[36] as compared with urine, which requires adjustment for hydration status with urinary creatinine or specific gravity[37–40] and expression of As metabolites as percentages rather than concentrations. Furthermore, intuitively, As metabolite concentrations in blood should more closely reflect As metabolite exposures to tissues as compared with urine where concentrations of As metabolites are much higher. The bAs methylation indices, PMI and SMI, have the advantage that they are less impacted by As exposure levels and water filter compliance, as observed in the present study, than are bAs metabolite concentrations.

In the first phase of FACT, we observed significant increases in As methylation capacity as reflected by greater decreases in bMMAs and PMI and increases in bDMAs and SMI in all treatment groups receiving FA relative to PBO. In addition, treatment effects on increasing SMI were seen as early as the first week of intervention for the creatine + 400FA treatment group. The nonsignificant increase in SMI from baseline to week 1 in the PBO group [2.1% (95% CI: −0.8, 5.2)] was likely due to water filter use,[22] given that water As concentrations have been negatively associated with SMI in urine[33] and that there is experimental evidence showing that arsenic methylation capacity is saturable.[41] Although no significant changes in bInAs concentrations were observed for any treatment groups compared with PBO over time, this was likely due to its methylation to MMAs and continued InAs exposure owing to poor compliance with As-removal filters.[22] In exploratory sex-stratified analyses, FA supplementation results were similar among male and female participants, whereas creatine supplementation alone resulted in significantly greater decreases in bMMAs and increases in SMI compared with PBO only among male participants. Although the changes in bMMAs and SMI with creatine treatment compared with PBO were not significant in female participants, they were in the same direction as those in male participants. Thus, given that our study was not powered to detect effect modification, this could be an issue of small sample size, or it could possibly be related to differences in creatine metabolism due to higher muscle mass among male participants. The finding that FA supplementation was effective in increasing As methylation capacity in this population that was predominantly folate sufficient (plasma folate was $\geq 9.0\,\text{nmol/L}$ in 80.3% of the study participants) has important implications. For example, if FA supplementation was only effective in folate-deficient individuals, then targeted approaches would be more appropriate than widespread food FA fortification programs. The latter can influence entire populations and overcome problems associated with FA supplementation, such as poor long-term compliance and difficulties in reaching vulnerable subsets of the population.

In the second phase of FACT, the data showed evidence of the reversal of treatment effects on As methylation following FA cessation from weeks 12 to 24, with significant decreases in SMI and bDMAs in those who switched from 800FA to PBO, whereas for those who remained on 800FA, PMI and bMMAs concentrations continued to decline, showing added benefit to a longer intervention. In addition, as compared with baseline, there was a significantly greater increase in SMI at week 24 among participants who continued supplementation as compared with those who switched to PBO. Data on bDMAs concentrations also suggest the reversal of treatment effects on As methylation after FA cessation: the 7.77% increase in bDMAs concentrations between baseline and week 24 among participants who discontinued 400FA supplementation was roughly half that of those who continued FA supplementation (15.6%). These data indicate that sustainable improvement in As methylation capacity may require long-term interventions, such as food fortification programs, which are generally low-cost, effective approaches that can achieve broad coverage.

The mechanisms of As toxicity are multifactorial; for example, chromosomal instability and altered methylation of DNA and histones are two mechanisms involved in As toxicity, as reviewed in Abuawad et al.[42] These mechanisms can be exacerbated by impairments in OCM and may also be improved by folate supplementation. There is some evidence to suggest that folate and/or other OCM-related micronutrients may also be beneficial in reducing As-related toxicity through these mechanisms, in addition to our observed effects of increasing As methylation and lowering bAs, although additional data are needed.[43,44]

The creatine treatment arms were designed to test our *a priori* hypothesis that creatine supplementation would reduce the high demand for SAM needed to support endogenous creatine biosynthesis, thereby sparing SAM for the methylation of As. This hypothesis is supported by positive associations between urinary creatinine levels and As methylation capacity in observational studies.[17,19,45–47] In FACT, 3 g creatine supplementation decreased creatine biosynthesis as indicated by significant decreases in plasma GAA, the precursor to creatine.[48] In our previous FACT report on urinary As metabolites, the group receiving creatine supplementation had a significantly greater decrease in %MMAs in urine compared with PBO following treatment, but no significant creatine treatment effects were observed for %InAs or %DMAs.[18] In the present analyses, the group receiving creatine supplementation alone had a significantly greater decrease in bMMAs concentrations and increase in SMI from baseline to week 12 compared with PBO. However, the creatine treatment effect was small compared with that of FA supplementation, and there were no significant differences observed in changes of bInAs, bDMAs, or PMI. Potential explanations for limited creatine treatment effects include long-range allosteric regulation of OCM that regulate intracellular SAM concentrations, for example, by changes in activity of glycine-*N*-methyltransferase (GNMT), a major regulator of SAM concentrations, and in flux through the transsulfuration pathway, which is stimulated by SAM.[49–51] The strong cross-sectional associations previously reported between urinary creatinine and As methylation patterns may partially be related to dietary creatine intake, but they could also potentially be influenced by dietary protein/methionine content associated with meat intake or other factors, such as renal function, which could differentially influence excretion/reabsorption of As metabolites in urine.

A limitation of this study includes decreased compliance with water filter use over time, as reported by Sanchez et al. in a study of total urine As and water As levels in FACT,[22] which complicates interpretation of changes in bAs metabolite concentrations given that there was a strong decline in bAs for all groups at week 1 owing to initial compliance that reverted afterward. However, owing to the randomized study design, there is no evidence to suggest that compliance differed between treatment groups, which is supported by our data on total urinary As concentrations. Thus, this is not expected to change the conclusions of the study. Furthermore, the use of the As methylation indices, PMI and SMI, circumvents this issue that more directly affects bAs metabolite concentrations. Strengths of this study include the double-blind, randomized, PBO-controlled study design, including evaluation of the potential reversal of treatment effects on As metabolites following cessation of FA treatment and the high participant compliance with treatment. In addition, the use of blood, rather than urinary, measures of As metabolites may better reflect concentrations in tissues and be less affected by hydration levels. That a large proportion of the study population were folate replete increases generalizability to other As-exposed populations.

Conclusion

Our results indicate that FA supplementation increased As methylation capacity, as reflected by larger decreases in PMI and MMAs and increases in SMI and DMAs in blood within FA-treated groups compared with PBO. These findings are consistent with our previous report of the effects of FA supplementation on urinary As methylation profiles in this trial.[18] In recent years, the evidence supporting the critical role of nutritional factors in As methylation and corresponding toxicity has grown substantially.[42,52] Although As-remediation programs aimed at reducing exposure to As through contaminated drinking water should always remain the top priority, additional approaches to decrease the chronic health effects of As exposure through increasing As methylation capacity and lowering bAs concentrations are needed.

Food fortification programs are one option. Approximately 85 countries mandate grain fortification with FA, but 28 European countries and most Asian countries do not or have only voluntary programs.[53] A major goal of FA fortification is to reduce neural tube defects, a lifelong debilitating condition due to failure of the neural folds to fuse during embryonic development. Fortification circumvents issues associated with FA supplementation or other dietary public health interventions, such as unplanned pregnancies, poor supplementation compliance, and difficulties reaching segments of the population.[54] However, despite the many known benefits of FA fortification, there are knowledge gaps regarding the potential risks of excessive folate intake and/or high FA levels, particularly in populations where use of folate supplements is common. These are summarized in the proceedings of a National Institutes of Health (NIH) workshop and include the masking of B_{12} deficiency, metabolic interactions with B_{12} that may influence neurocognitive outcomes among the elderly, and potential adverse effects on cancer risk, birth outcomes, and other diseases.[55] In addition, FA is a synthetic form of folate used in food fortification because it is chemically more stable than 5-mTHF, which is easily degraded by prolonged cooking. Individuals vary in their ability to fully metabolize FA to 5-mTHF. As a result, detectable levels of circulating unmetabolized FA are nearly ubiquitous in populations with FA fortification programs, such as in the United States.[56] Evidence of potential unanticipated effects of unmetabolized FA is controversial and is described by the NIH workshop proceedings as inconclusive.[55]

Many parts of the world affected by high levels of As contamination of drinking water, such as Bangladesh, do not currently have voluntary, let alone mandatory, FA fortification programs, and folate deficiency is common.[57] Carefully executed FA fortification programs or other public health interventions are needed to eradicate folate deficiency and its associated adverse health outcomes and could also facilitate a partial reduction in the extensive public health burden of As exposure.

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